

QUESTION 23

Context-Aware Chatbot Using LangChain, Retrieval and an LLM

Domain: Artificial Intelligence and Machine Learning

Framework: LangChain | Vector Database: FAISS | LLM: FLAN-T5

1. Aim

To implement a context-aware chatbot that combines LangChain retrieval with conversation history and a language model so that the system can maintain context across multiple questions in a single session.

2. Problem Statement

A chatbot that treats every question independently may struggle with follow-up questions containing references such as “its”, “they”, or “which one”. A context-aware system stores previous exchanges and supplies them along with retrieved knowledge to the language model.

3. Objectives

- Create an AI and Machine Learning domain knowledge base.
- Store semantic embeddings in a FAISS vector database.
- Use a LangChain retriever to obtain relevant knowledge.
- Maintain conversation history during the current session.
- Combine previous conversation with retrieved knowledge.
- Generate responses using an LLM.
- Demonstrate multi-turn chatbot interaction.

4. System Architecture

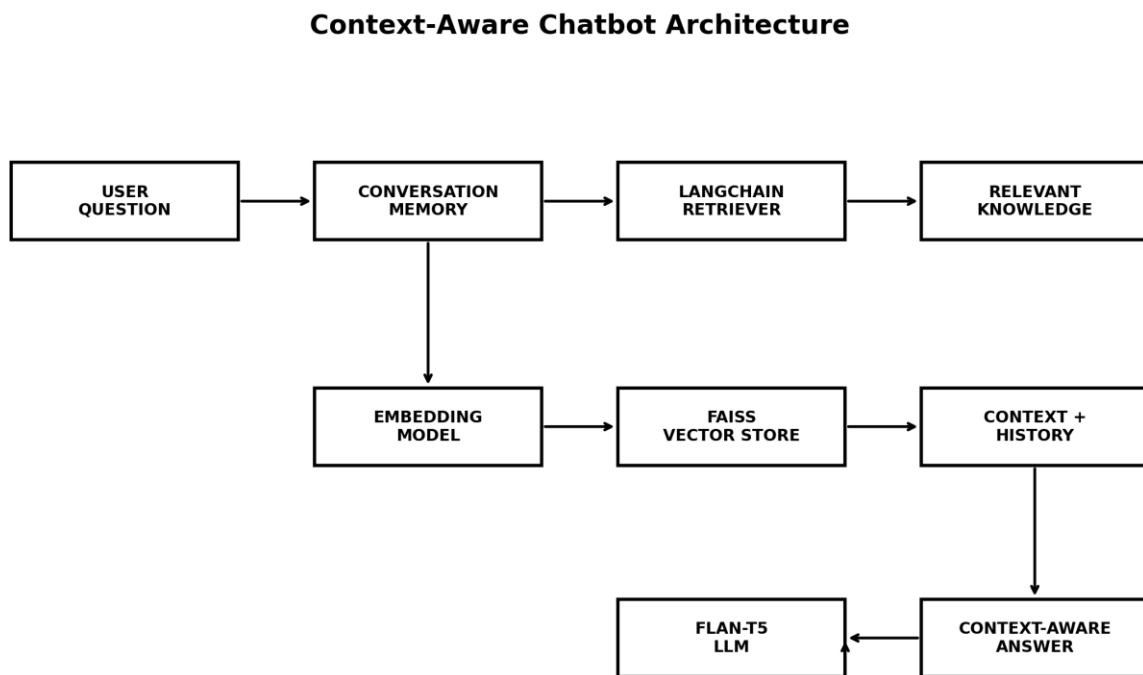


Figure 1: Context-aware chatbot architecture.

5. Working Principle

For each user question, the system retrieves relevant domain documents through LangChain and FAISS. It also checks the stored conversation history. The retrieved knowledge and previous exchanges are combined into a prompt and sent to the FLAN-T5 model. After generating the answer, the new question-answer pair is stored in conversation history for subsequent turns.

6. Technology Stack

Component	Implementation
Programming Language	Python
Framework	LangChain
Embedding Model	all-MiniLM-L6-v2
Vector Database	FAISS
Retriever	LangChain Retriever
Generation Model	google/flan-t5-small
Memory	Conversation history list
Domain	Artificial Intelligence and Machine Learning

7. Multi-Turn Test Procedure

1. Ask “What is machine learning?” to establish the initial topic.
2. Ask “What are its types?” as a follow-up question.
3. Ask “Which one uses labelled data?” to test another follow-up.
4. Ask “What is NLP?” to verify retrieval of a new topic.
5. Observe whether conversation context remains active after the first turn.

8. Actual Multi-Turn Chatbot Session

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CONTEXT-AWARE CHATBOT SESSION

You: What is machine learning?

Conversation Context: NEW SESSION
Retrieved Topics:
1. Machine Learning
2. Supervised Learning
3. Deep Learning
Chatbot:
a branch of artificial intelligence that enables computers to learn patterns from data
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You: What are its types?

Conversation Context: ACTIVE
Previous exchanges: 1
Retrieved Topics:
1. Computer Vision
2. Deep Learning
3. Unsupervised Learning
Chatbot:
Deep learning
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You: Which one uses labelled data?

Conversation Context: ACTIVE
Previous exchanges: 2
Retrieved Topics:
1. Supervised Learning
2. Unsupervised Learning
3. Machine Learning
Chatbot:
Deep learning
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You: What is NLP?

Conversation Context: ACTIVE
Previous exchanges: 3
Retrieved Topics:
1. NLP
2. Supervised Learning
3. Reinforcement Learning
Chatbot:
Natural language processing
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Figure 2: Actual multi-turn session output from the implementation.

9. Conversation Results

Turn	Question	Top Retrieved Topic	Chatbot Response
1	What is machine learning?	Machine Learning	a branch of artificial intelligence that enables computers to learn patterns from data
2	What are its types?	Computer Vision	Deep learning
3	Which one uses labelled data?	Supervised Learning	Deep learning
4	What is NLP?	NLP	Natural language processing

10. Context Status Analysis

The first question started a new session. After that, the system reported “Conversation Context: ACTIVE” and tracked the number of previous exchanges: 1 before the second question, 2 before the third, and 3 before the fourth. This confirms that conversation history was being maintained across the session.

11. Retrieval Analysis

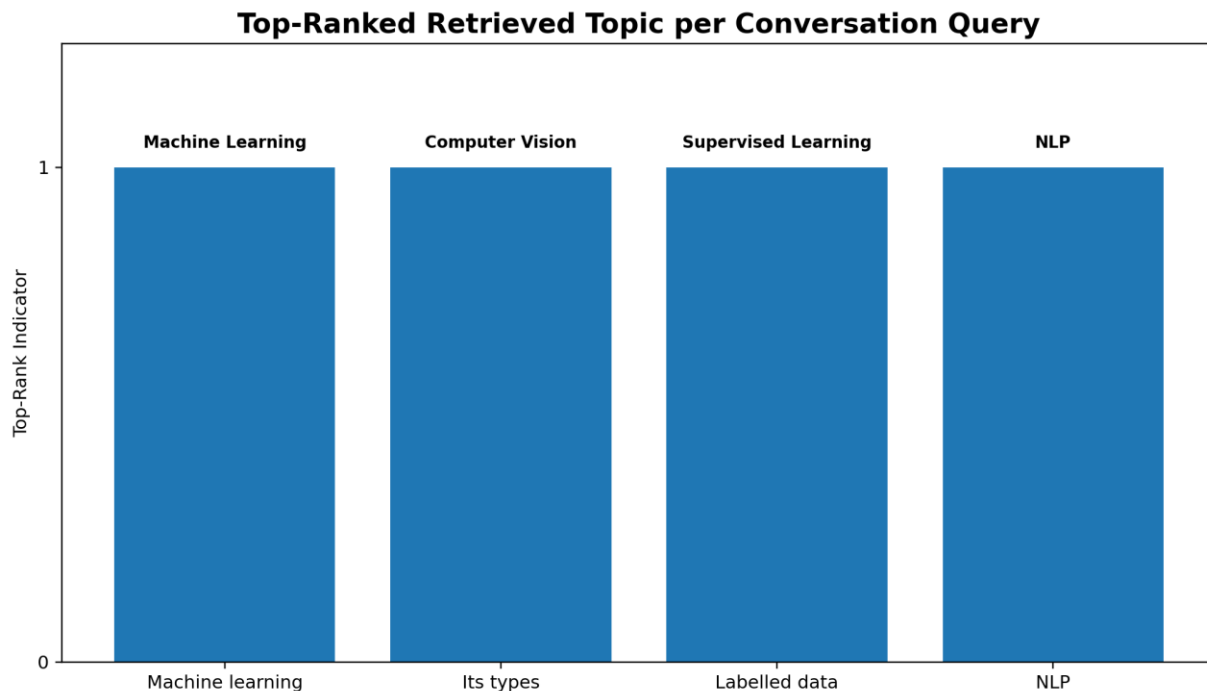


Figure 3: Top-ranked retrieved topic for each tested question.

The first question retrieved Machine Learning as the top topic. The follow-up “What are its types?” returned Computer Vision as the first retrieved topic, showing that retrieval was not fully aligned with the conversational reference. The third question correctly retrieved Supervised Learning first. The fourth question correctly retrieved NLP first.

12. Response Quality Analysis

The chatbot produced a relevant machine-learning definition for the first question and correctly answered the NLP query with “Natural language processing.” However, the responses to the two follow-up questions were not fully accurate: “What are its types?” returned “Deep learning,” and “Which one uses labelled data?” also returned “Deep learning,” even though the knowledge base identifies supervised learning as the approach that uses labelled data. This is an important experimental observation: the conversation-memory mechanism is active, but the small FLAN-T5 model does not always resolve follow-up references correctly.

13. Result

The context-aware chatbot successfully maintained conversation history across four turns and reported active context after the first exchange. Retrieval correctly identified relevant topics for the first, third, and fourth queries, while the second query demonstrated a retrieval mismatch. The experiment therefore confirms the implementation of context tracking and retrieval, while also revealing limitations of the small generation model.

14. Advantages

- Maintains conversation history during a session.
- Combines retrieved knowledge with previous exchanges.
- Supports follow-up questions.
- Uses semantic retrieval rather than only keyword matching.
- Can be extended with larger language models for improved contextual reasoning.

15. Limitations Observed

- The small FLAN-T5 model sometimes produces overly short or incorrect follow-up answers.
- Conversational references such as “its” may not always be resolved correctly.
- Retriever ranking can be affected by ambiguous follow-up questions.
- The current memory is limited to the active program session.

16. Applications

- Educational conversational assistants
- Technical support chatbots
- Domain-specific help systems
- Knowledge-base assistants
- Interactive learning systems

17. Conclusion

A context-aware chatbot was successfully implemented using LangChain, FAISS retrieval, conversation history, and the FLAN-T5 language model. The system maintained active conversation context across four turns and successfully handled direct questions such as machine learning and NLP. The experiment also showed that context tracking alone does not guarantee correct follow-up reasoning: the small generation model produced incorrect answers for two follow-up questions. Thus, the implementation demonstrates both the working principle and practical limitations of a context-aware retrieval chatbot.